

10 Video lessons

Course Introduction

Video: Specification Overview

2:00

Video: Meet Your Instructor

1:00

Reading: Course Syllabus

15:00

Video: Summary of Course 2

1:00

Video: Course Overview

4:00

Video: Introduction to Our Learning Environment

2:00

Reading: Textbook: Setting up our Coding Environment - Python in Processing

15:00

Reading: Python Code for Creating Color for Designers (See Below)

15:00

Discussion: Prerequisites: Introduction to Python

15:00

Reading: Pre-Course Survey

15:00

Introduction to Particle Systems

Working with Forces

Search in course

Search

Textbook: Setting up our Coding Environment - Python in Processing

This is a reminder of courses 1 & 2. Please skip this lesson if you have completed these courses and already have Processing installed.

Here we will learn how to install and set up Processing 3.5.4 with Python mode, enabling them to write and visualize Python code in an interactive environment.

Choosing our learning environment

Processing is a flexible software stack (think of it as a language for learning how to code within the context of the visual arts, developed by Casey Reas and Ben Fry), was initially created to serve as a foundation for teaching computer programming the fundamental of computer programming in a visual context. The essence of Processing lies in its ability to not just create, making it a great platform for artists, designers, architects, and anyone interested in creating digital art or interactive projects.

While **Processing 3** is the latest version and offers a range of updated features and improvements, we have chosen to use **Processing 3.5.4** for our course, particularly for those working in **Python mode**.

The decision stems from the fact that Processing 4 Python mode currently has some bugs that can significantly impact the learning experience. One notable issue is the environment's inability to highlight errors in Python code effectively. This feature is crucial for beginners, as it helps them quickly identify and understand mistakes in their code. Unfortunately, in Processing 4, the error-highlighting functionality is not working as reliably as it does in Processing 3.5.4. To ensure that all students have a smooth and educational coding experience where they can easily see and correct their mistakes, we will be working with Processing 3.5.4. This version provides a stable and supportive environment for learning Python in a dynamic, visual context.

Now, let's install Processing.

- Step 1: Download Processing 3.5.4
 - Go to the [Processing download site](#).

- Select the version compatible with their operating system under the "list of stable releases" section, specifically **Processing 3.5.4**.

- The download link will allow you to download the zip file package that contains Processing. Once you download it, unzip it and you should see the folder name:

- You can execute processing by just executing the Processing.exe file. No installation will be required. We suggest that you keep track of this folder and place it somewhere, such as your Program Files folder.

Step 2: Install Python Mode in Processing

- Open Processing:** Find the Processing application on your computer and open it.
- Access Modes:** In the Processing application, look at the top right corner for a dropdown menu that says "Modes". Click on it and then select "Python Mode 3.7".

- Switch to Python Mode:** Click the "Mode Manager" window. Click again on this mode dropdown menu at the top right (bright off say "Java") and select "Python Mode 3.7".

- Install Python Mode:** In the "Mode Manager" window that appears, scroll down to find "Python Mode for Processing 3.7" and click the "Install" button next to it. Wait for the installation to complete.

You can also check out the [documentation and tutorials](#) for Processing Python.

Get to next item

Completed

Like Dislike Report an issue

English 1

14